

Supplementary Material for “Multi-Semivalue Ranking via Shared Utility Evaluation”

Chirong Zhang
Simon Fraser University
Vancouver, BC, Canada
chirong_zhang@sfu.ca

Xiaohui Yu
York University
Toronto, ON, Canada
xhyu@yorku.ca

Xuan Luo
York University
Toronto, ON, Canada
xuanluo@yorku.ca

Jian Pei
Duke University
Durham, NC, USA
j.pei@duke.edu

Supplementary Material Catalog. The entries and page numbers below link directly to the corresponding content.

Content	Page
A Notation	p. 2
B Algorithm	p. 2
B.1 Boundary-exact ExcRank Algorithm	p. 2
C Proofs	p. 3
C.1 ExcRank Rank Preservation	p. 3
C.2 Closed-form Semivalues for SOUG	p. 3
C.3 Concentration and Ranking-Recovery Proofs	p. 3
• IncRank Ranking Recovery	p. 4
• ExcRank Ranking Recovery	p. 4
• Fixed-Weight Combination	p. 4
• AdaRank Ranking Recovery	p. 5
D Experimental Details	p. 5
• Machine Configuration	p. 5
• Dataset and Utility Construction	p. 5
• Reference-Ranking Creation	p. 5
E Additional Experimental Results	p. 5
E.1 Sampling-Distribution Sensitivity	p. 5
E.2 Top-Selection Diagnostics	p. 6
E.3 Single-Target Ranking Results	p. 6
E.4 Adaptive Weight Selection	p. 7
E.5 Runtime Overhead	p. 8
E.6 CIFAR-10N Audit-Size Sensitivity	p. 9
E.7 Pairwise Inversion Decay	p. 9

A Notation

Table 1: Notation used in the paper

Notation	Definition
n	the number of players
K	the number of target semivalues
b	the per-player utility-evaluation budget
B	the total utility-evaluation budget, $B = bn$
T	the interior sampling budget
$\mathcal{N}$	the player set $\{x_1, \dots, x_n\}$
x_i	the i -th player
$v(\cdot)$	the utility function
S	a coalition
ϕ_i	the semivalue of player x_i
$\mathbf{p}$	the semivalue weight vector $\langle p_1, \dots, p_n \rangle$
Φ	the target set of semivalues
$\pi^{(j)}$	the ranking induced by semivalue $\phi^{(j)}$
$\mathcal{D}_i^-$	$\{S \subseteq \mathcal{N} \mid x_i \notin S, 0 < S < n\}$
$\mathcal{D}_{i,s}^-$	$\{S \in \mathcal{D}_i^- \mid S = s\}$
$\mathcal{D}_i^+$	$\{S \subseteq \mathcal{N} \mid x_i \in S, 0 < S < n\}$
$\mathcal{D}_{i,s}^+$	$\{S \in \mathcal{D}_i^+ \mid S = s\}$
$\mathcal{D}_{\cdot,s}$	$\{S \subseteq \mathcal{N} \mid S = s\}$
Y_i^+, Y_i^-	include-side and exclude-side ranking proxies
q_s	the sampling probability for coalition size s
$\mathbb{P}$	a probability measure

B Algorithm

B.1 Boundary-exact ExcRank Algorithm

Algorithm 1: ExcRank for one target semivalue

Input: players $\mathcal{N}$, budget $B \geq 2n$, semivalue weights $\mathbf{p}$, size probabilities $\{q_s\}_{s=2}^{n-2}$

Output: ExcRank scores $\widehat{Y}_i^-$ for all $i \in \mathcal{N}$

- 1 $\widehat{Y}_{i,s}^-, c_{i,s}^- \leftarrow 0$ for all $i \in \mathcal{N}$ and $1 \leq s \leq n-1$;
- 2 evaluate $u_i^{(1)} \leftarrow v(\{i\})$ and $u_i^{(n-1)} \leftarrow v(\mathcal{N} \setminus \{i\})$ for all $i \in \mathcal{N}$;
- 3 $\widehat{Y}_{i,1}^- \leftarrow \frac{1}{n-1} \sum_{j \neq i} u_j^{(1)}$ and $\widehat{Y}_{i,n-1}^- \leftarrow u_i^{(n-1)}$ for all $i \in \mathcal{N}$;
- 4 **for** $t = 1, \dots, B - 2n$ **do**
- 5 sample a coalition size $s \in \{2, \dots, n-2\}$ with probabilities $\{q_s\}_{s=2}^{n-2}$;
- 6 sample a coalition S_t uniformly from $\mathcal{D}_{\cdot,s}$;
- 7 evaluate $v(S_t)$;
- 8 **for each** $i \notin S_t$ **do**
- 9 $\widehat{Y}_{i,s}^- \leftarrow \frac{c_{i,s}^- \widehat{Y}_{i,s}^- + v(S_t)}{c_{i,s}^- + 1}$;
- 10 $c_{i,s}^- \leftarrow c_{i,s}^- + 1$;
- 11 $\widehat{Y}_i^- \leftarrow -\sum_{s=1}^{n-1} (p_s + p_{s+1}) \binom{n-1}{s} \widehat{Y}_{i,s}^-$ for all $i \in \mathcal{N}$;
- 12 **return** $\widehat{Y}_i^-$ for all $i \in \mathcal{N}$

Algorithm 1 summarizes the complete **ExcRank** procedure. The boundary strata ($s = 1$ and $s = n - 1$) contain $2n$ coalitions in total and are evaluated exactly, while the remaining $B - 2n$ budget is allocated to the interior strata according to $\{q_s\}_{s=2}^{n-2}$. Each sampled coalition updates the corresponding size-conditioned averages for every player it does not contain, after which the final **ExcRank** scores are computed using $\widehat{Y}_i^- = -\sum_{s=1}^{n-1} (p_s + p_{s+1}) \binom{n-1}{s} \widehat{Y}_{i,s}^-$ for all $i \in \mathcal{N}$. Since the target semivalue affects only the final aggregation coefficients, the same sampled coalition utilities can be reused to estimate the rankings of multiple semivalues without additional utility evaluations.

C Proofs

C.1 ExcRank Rank Preservation

LEMMA 1 (**EXCRANK** RANK PRESERVATION). *For any semivalue with weight vector $\mathbf{p} = \langle p_1, \dots, p_n \rangle$, there exists a constant*

$$C^- = \sum_{\emptyset \neq S \subseteq N} p_{|S|} v(S) - p_1 v(\emptyset),$$

*independent of the player i , such that $\phi_i = Y_i^- + C^-$ for every $i \in N$. Consequently, $Y_j^- - Y_k^- = \phi_j - \phi_k$ for any two players $j, k \in N$, and **ExcRank** induces exactly the same ranking as the original semivalue.*

PROOF. Starting from the semivalue definition and writing $s = |S|$, we have

$$\begin{aligned} \phi_i &= \sum_{S \subseteq N, i \in S} p_s v(S) - \sum_{S \subseteq N \setminus \{i\}} p_{s+1} v(S) \\ &= \sum_{\emptyset \neq S \subseteq N} p_s v(S) - \sum_{\emptyset \neq S \subseteq N \setminus \{i\}} p_s v(S) \\ &\quad - p_1 v(\emptyset) - \sum_{\emptyset \neq S \subseteq N \setminus \{i\}} p_{s+1} v(S) \\ &= - \sum_{S \in \mathcal{D}_i^-} (p_s + p_{s+1}) v(S) + \sum_{\emptyset \neq S \subseteq N} p_s v(S) - p_1 v(\emptyset) \\ &= Y_i^- + C^-, \end{aligned}$$

where

$$C^- = \sum_{\emptyset \neq S \subseteq N} p_s v(S) - p_1 v(\emptyset)$$

is independent of the player i . Hence, for any two players $j, k \in N$,

$$\phi_j - \phi_k = (Y_j^- + C^-) - (Y_k^- + C^-) = Y_j^- - Y_k^-.$$

Therefore, **ExcRank** preserves every pairwise semivalue gap and thus induces exactly the same ranking as the original semivalue. $\square$

C.2 Closed-form Semivalues for SOUG

Consider the SOUG benchmark used in the experiments, with unanimity sets $A_1, \dots, A_M \subseteq N$ and nonnegative weights $c_1, \dots, c_M \geq 0$:

$$v(S) = \sum_{m=1}^M c_m \mathbf{1}\{A_m \subseteq S\}.$$

For a semivalue with weights $\mathbf{p} = \langle p_1, \dots, p_n \rangle$, the exact value of player i is

$$\phi_i(\mathbf{p}) = \sum_{m: i \in A_m} c_m \sum_{r=|A_m|}^n \binom{n-|A_m|}{r-|A_m|} p_r.$$

To prove this formula, first consider one unanimity game $u_A(S) = \mathbf{1}\{A \subseteq S\}$. If $i \notin A$, then adding i cannot complete the unanimity set, so every marginal contribution is zero. If $i \in A$, then

$$u_A(S \cup \{i\}) - u_A(S) = 1$$

exactly when $A \setminus \{i\} \subseteq S$, where $S \subseteq N \setminus \{i\}$. For coalitions of size $r-1$, the number of such coalitions is $\binom{n-|A|}{r-|A|}$, because the remaining $r-|A|$ players can be chosen from $N \setminus A$. Therefore, this unanimity game contributes

$$\mathbf{1}\{i \in A\} \sum_{r=|A|}^n \binom{n-|A|}{r-|A|} p_r$$

to $\phi_i(\mathbf{p})$. Semivalues are linear in the utility function, so the same argument applies to weighted sums of unanimity games. Summing the weighted unanimity-game values over all m gives the stated closed form for SOUG.

C.3 Proof of the Ranking Error Results

The ranking analysis has two steps. First, we control the sampling error of a generic linear combination of include-side and exclude-side stratum averages. Then each theorem statement follows by choosing the appropriate coefficients and applying a pairwise union bound. Boundary strata are evaluated exactly, so they appear in both the exact score and the estimated score with the same value and contribute no sampling error.

LEMMA 2 (GENERIC PER-PLAYER CONCENTRATION). *Fix a player i . For $s \in \mathcal{S}_{\text{rand}}$, let*

$$Z_i = \sum_{s \in \mathcal{S}_{\text{rand}}} (a_s^+ Y_{i,s}^+ + a_s^- Y_{i,s}^-), \quad \widehat{Z}_i = \sum_{s \in \mathcal{S}_{\text{rand}}} (a_s^+ \widehat{Y}_{i,s}^+ + a_s^- \widehat{Y}_{i,s}^-),$$

where the coefficients a_s^+ and a_s^- are fixed and not all zero. Let

$$D_a(\mathbf{q}) = \sum_{s \in \mathcal{S}_{\text{rand}}} \frac{n}{q_s} \left(\frac{(a_s^+)^2}{s} + \frac{(a_s^-)^2}{n-s} \right).$$

Let $\gamma_a(\mathbf{q})$ be the minimum expected sampling fraction over the active side-size pairs,

$$\gamma_a(\mathbf{q}) = \min \left(\left\{ \frac{q_s s}{n} : a_s^+ \neq 0 \right\} \cup \left\{ \frac{q_s (n-s)}{n} : a_s^- \neq 0 \right\} \right),$$

and let

$$h(a) = \mathbf{1}\{\exists s : a_s^+ \neq 0\} + \mathbf{1}\{\exists s : a_s^- \neq 0\}$$

be the number of active side families. If $\|\mathbf{v}\|_\infty \leq u$, then for any $\epsilon > 0$,

$$\mathbb{P}(|\widehat{Z}_i - Z_i| \geq \epsilon) \leq 4h(a)n \exp \left(-T \min \left\{ \frac{\epsilon^2}{4D_a(\mathbf{q})u^2}, \frac{\gamma_a(\mathbf{q})^2}{2} \right\} \right).$$

PROOF. Let $\mathcal{S}_1, \dots, \mathcal{S}_T$ be independent interior coalition samples: first draw a size $s \in \mathcal{S}_{\text{rand}}$ with probability q_s , then draw $\mathcal{S}_t$ uniformly among coalitions of size s . Define the include-side and exclude-side counts

$$T_{i,s}^+ = \sum_{t=1}^T \mathbf{1}\{i \in \mathcal{S}_t, |\mathcal{S}_t| = s\}, \quad T_{i,s}^- = \sum_{t=1}^T \mathbf{1}\{i \notin \mathcal{S}_t, |\mathcal{S}_t| = s\}.$$

Let $r_{i,s}^\pm = T_{i,s}^\pm / T$. Since the coalition size is sampled with probability q_s and the coalition is uniform conditional on size,

$$\tau_{i,s}^+ = \mathbb{E}[r_{i,s}^+] = \frac{q_s s}{n}, \quad \tau_{i,s}^- = \mathbb{E}[r_{i,s}^-] = \frac{q_s (n-s)}{n}.$$

For $\omega > 0$, let $G_i(a, \omega)$ be the event that $r_{i,s}^\pm \geq \tau_{i,s}^\pm - \omega$ for every active side-size pair. Hoeffding's inequality and a union bound over the lower tails give

$$\mathbb{P}(G_i(a, \omega)^c) \leq h(a)|\mathcal{S}_{\text{rand}}| \exp(-2T\omega^2).$$

If $\omega \leq \gamma_a(\mathbf{q})/2$, then on $G_i(a, \omega)$ every active count is nonzero and at least half its expectation.

Condition on the category sequence $\{(|\mathcal{S}_t|, \mathbf{1}\{i \in \mathcal{S}_t\})\}_{t=1}^T$ and on a realization in $G_i(a, \omega)$. The coalitions contributing to each active side-size pair are independent uniform draws from the corresponding stratum. Zero-coefficient terms are omitted from the

inverse-count sums below. For the one-sided event $Z_i - \widehat{Z}_i \geq \epsilon$, Chernoff's method and Hoeffding's lemma give

$$\begin{aligned} & \mathbb{P}\left(Z_i - \widehat{Z}_i \geq \epsilon \mid \text{category sequence}\right) \\ & \leq \inf_{\lambda > 0} \exp\left(\frac{\lambda^2 u^2}{2} \sum_{s \in \mathcal{S}_{\text{rand}}} \left[\frac{(a_s^+)^2}{T_{i,s}^+} + \frac{(a_s^-)^2}{T_{i,s}^-}\right] - \lambda \epsilon\right) \\ & \leq \inf_{\lambda > 0} \exp\left(\frac{\lambda^2 u^2}{T} D_a(\mathbf{q}) - \lambda \epsilon\right). \end{aligned}$$

Optimizing over λ yields

$$\mathbb{P}\left(Z_i - \widehat{Z}_i \geq \epsilon \mid \text{category sequence}\right) \leq \exp\left(-\frac{T \epsilon^2}{4 D_a(\mathbf{q}) u^2}\right).$$

Combining this bound with the bad-count event and choosing

$$\omega = \min\left\{\frac{\epsilon}{\sqrt{8 D_a(\mathbf{q}) u^2}}, \frac{\gamma_a(\mathbf{q})}{2}\right\}$$

gives

$$\mathbb{P}(Z_i - \widehat{Z}_i \geq \epsilon) \leq 2h(a)n \exp\left(-T \min\left\{\frac{\epsilon^2}{4 D_a(\mathbf{q}) u^2}, \frac{\gamma_a(\mathbf{q})^2}{2}\right\}\right),$$

where we use $|\mathcal{S}_{\text{rand}}| \leq n$ and $h(a)n \geq 1$. Applying the same argument to $-a_s^+$ and $-a_s^-$ controls the opposite one-sided event. Adding the two tails proves the lemma. $\square$

We now restate the main ranking-recovery guarantees for completeness.

THEOREM 1 (INCRANK RANKING RECOVERY). *Assume $\|v\|_\infty \leq u$. For two players j, k , let $\Delta_{jk}^+ = Y_j^+ - Y_k^+ = \phi_j - \phi_k$. If $\Delta_{jk}^+ > 0$, then*

$$\begin{aligned} & \mathbb{P}\left((Y_j^+ - Y_k^+)(\widehat{Y}_j^+ - \widehat{Y}_k^+) < 0\right) \\ & \leq 8n \exp\left(-T \min\left\{\frac{(\Delta_{jk}^+)^2}{16 D^+(\mathbf{q}) u^2}, \frac{\gamma^+(\mathbf{q})^2}{2}\right\}\right), \end{aligned}$$

where

$$\gamma^+(\mathbf{q}) = \min_{s \in \mathcal{S}_{\text{rand}}} \frac{q_s s}{n}, \quad m_s^+ = (p_s + p_{s+1}) \binom{n-1}{s-1},$$

and

$$D^+(\mathbf{q}) = \sum_{s \in \mathcal{S}_{\text{rand}}} \frac{n}{q_s} \frac{(m_s^+)^2}{s}.$$

PROOF. Apply Lemma 2 with

$$a_s^+ = m_s^+ = (p_s + p_{s+1}) \binom{n-1}{s-1}, \quad a_s^- = 0.$$

Then $h(a) = 1$, $\gamma_a(\mathbf{q}) \geq \gamma^+(\mathbf{q})$, and $D_a(\mathbf{q}) = D^+(\mathbf{q})$. Therefore, for each player i ,

$$\mathbb{P}\left(|\widehat{Y}_i^+ - Y_i^+| \geq \epsilon\right) \leq 4n \exp\left(-T \min\left\{\frac{\epsilon^2}{4 D^+(\mathbf{q}) u^2}, \frac{\gamma^+(\mathbf{q})^2}{2}\right\}\right).$$

If $Y_j^+ > Y_k^+$, a ranking inversion implies that at least one of the two estimation errors is at least $\Delta_{jk}^+/2$. Thus,

$$\begin{aligned} & \mathbb{P}\left((Y_j^+ - Y_k^+)(\widehat{Y}_j^+ - \widehat{Y}_k^+) < 0\right) \\ & \leq \mathbb{P}\left(|\widehat{Y}_j^+ - Y_j^+| \geq \frac{\Delta_{jk}^+}{2}\right) + \mathbb{P}\left(|\widehat{Y}_k^+ - Y_k^+| \geq \frac{\Delta_{jk}^+}{2}\right) \\ & \leq 8n \exp\left(-T \min\left\{\frac{(\Delta_{jk}^+)^2}{16 D^+(\mathbf{q}) u^2}, \frac{\gamma^+(\mathbf{q})^2}{2}\right\}\right). \end{aligned}$$

$\square$

COROLLARY 1 (EXCRANK RANKING RECOVERY). *Assume $\|v\|_\infty \leq u$. For two players j, k , let $\Delta_{jk}^- = Y_j^- - Y_k^- = \phi_j - \phi_k$. If $\Delta_{jk}^- > 0$, then*

$$\begin{aligned} & \mathbb{P}\left((Y_j^- - Y_k^-)(\widehat{Y}_j^- - \widehat{Y}_k^-) < 0\right) \\ & \leq 8n \exp\left(-T \min\left\{\frac{(\Delta_{jk}^-)^2}{16 D^-(\mathbf{q}) u^2}, \frac{\gamma^-(\mathbf{q})^2}{2}\right\}\right), \end{aligned}$$

where

$$\gamma^-(\mathbf{q}) = \min_{s \in \mathcal{S}_{\text{rand}}} \frac{q_s(n-s)}{n}, \quad m_s^- = (p_s + p_{s+1}) \binom{n-1}{s},$$

and

$$D^-(\mathbf{q}) = \sum_{s \in \mathcal{S}_{\text{rand}}} \frac{n}{q_s} \frac{(m_s^-)^2}{n-s}.$$

PROOF. Apply Lemma 2 with

$$a_s^+ = 0, \quad a_s^- = -m_s^- = -(p_s + p_{s+1}) \binom{n-1}{s}.$$

The sign is irrelevant in the concentration bound because the coefficients enter through their squares. Hence $h(a) = 1$, $\gamma_a(\mathbf{q}) \geq \gamma^-(\mathbf{q})$, and $D_a(\mathbf{q}) = D^-(\mathbf{q})$. The same pairwise union bound as above gives

$$\begin{aligned} & \mathbb{P}\left((Y_j^- - Y_k^-)(\widehat{Y}_j^- - \widehat{Y}_k^-) < 0\right) \\ & \leq 8n \exp\left(-T \min\left\{\frac{(\Delta_{jk}^-)^2}{16 D^-(\mathbf{q}) u^2}, \frac{\gamma^-(\mathbf{q})^2}{2}\right\}\right). \end{aligned}$$

$\square$

COROLLARY 2 (FIXED-WEIGHT COMBINATION). *Assume $\|v\|_\infty \leq u$. For fixed constants $\mu^+, \mu^- \in \mathbb{R}$, fixed positive constants $\sigma^+, \sigma^- > 0$, and a fixed $w \in [0, 1]$, define*

$$Y_i(w) = w \frac{Y_i^+ - \mu^+}{\sigma^+} + (1-w) \frac{Y_i^- - \mu^-}{\sigma^-},$$

and define $\widehat{Y}_i(w)$ analogously from $\widehat{Y}_i^+$ and $\widehat{Y}_i^-$. For two players j, k , let $\Delta_{jk} = \phi_j - \phi_k > 0$. Then $\Delta_{jk}(w) = Y_j(w) - Y_k(w) > 0$, and

$$\begin{aligned} & \mathbb{P}\left((Y_j(w) - Y_k(w))(\widehat{Y}_j(w) - \widehat{Y}_k(w)) < 0\right) \\ & \leq 16n \exp\left(-T \min\left\{\frac{\Delta_{jk}(w)^2}{16 D_w(\mathbf{q}) u^2}, \frac{\bar{\gamma}(\mathbf{q})^2}{2}\right\}\right), \end{aligned}$$

where

$$\bar{\gamma}(\mathbf{q}) = \min\{\gamma^+(\mathbf{q}), \gamma^-(\mathbf{q})\},$$

and

$$D_w(\mathbf{q}) = \frac{w^2}{(\sigma^+)^2} D^+(\mathbf{q}) + \frac{(1-w)^2}{(\sigma^-)^2} D^-(\mathbf{q}).$$

PROOF. For fixed positive standardization constants, the centering constants do not affect the estimation error, so

$$\widehat{Y}_i(w) - Y_i(w) = \frac{w}{\sigma^+}(\widehat{Y}_i^+ - Y_i^+) + \frac{1-w}{\sigma^-}(\widehat{Y}_i^- - Y_i^-).$$

Apply Lemma 2 with

$$a_s^+ = \frac{w}{\sigma^+} m_s^+, \quad a_s^- = -\frac{1-w}{\sigma^-} m_s^-.$$

Then

$$D_a(\mathbf{q}) = \frac{w^2}{(\sigma^+)^2} D^+(\mathbf{q}) + \frac{(1-w)^2}{(\sigma^-)^2} D^-(\mathbf{q}) = D_w(\mathbf{q}).$$

Using $h(a) \leq 2$ and $\bar{y}(\mathbf{q}) = \min\{y^+(\mathbf{q}), y^-(\mathbf{q})\} \leq y_a(\mathbf{q})$ gives the per-player bound

$$\mathbb{P}\left(|\widehat{Y}_i(w) - Y_i(w)| \geq \epsilon\right) \leq 8n \exp\left(-T \min\left\{\frac{\epsilon^2}{4D_w(\mathbf{q})u^2}, \frac{\bar{y}(\mathbf{q})^2}{2}\right\}\right).$$

Since Y_i^+ and Y_i^- differ from ϕ_i only by player-independent constants, $\Delta_{jk} > 0$ implies positive include-side and exclude-side standardized pairwise gaps when $\sigma^+, \sigma^- > 0$. Their convex combination therefore satisfies $\Delta_{jk}(w) > 0$. Applying the pairwise union bound with $\epsilon = \Delta_{jk}(w)/2$ gives the stated bound. $\square$

COROLLARY 3 (**ADARANK RANKING RECOVERY**). *Consider the fixed-standardization version of **AdaRank**, where $\widehat{w} \in \mathcal{W}$ may be any data-dependent weight selected from a finite grid. Assume $\|v\|_\infty \leq u$ and $\Delta_{jk} = \phi_j - \phi_k > 0$. Then $\Delta_{jk}(w) = Y_j(w) - Y_k(w) > 0$ for every $w \in \mathcal{W}$, and*

$$\begin{aligned} &\mathbb{P}\left((Y_j(\widehat{w}) - Y_k(\widehat{w}))(\widehat{Y}_j(\widehat{w}) - \widehat{Y}_k(\widehat{w})) < 0\right) \\ &\leq 16n \sum_{w \in \mathcal{W}} \exp\left(-T \min\left\{\frac{\Delta_{jk}(w)^2}{16D_w(\mathbf{q})u^2}, \frac{\bar{y}(\mathbf{q})^2}{2}\right\}\right). \end{aligned}$$

In particular,

$$\begin{aligned} &\mathbb{P}\left((Y_j(\widehat{w}) - Y_k(\widehat{w}))(\widehat{Y}_j(\widehat{w}) - \widehat{Y}_k(\widehat{w})) < 0\right) \\ &\leq 16n|\mathcal{W}| \exp\left(-T \min_{w \in \mathcal{W}} \min\left\{\frac{\Delta_{jk}(w)^2}{16D_w(\mathbf{q})u^2}, \frac{\bar{y}(\mathbf{q})^2}{2}\right\}\right). \end{aligned}$$

PROOF. For **AdaRank**, $\widehat{w}$ is selected from a finite grid $\mathcal{W}$. By the same ordering-preservation argument, $\Delta_{jk} > 0$ implies $\Delta_{jk}(w) > 0$ for every $w \in \mathcal{W}$. For each realized value of $\widehat{w}$, a ranking inversion implies that the fixed- w inversion event occurs for at least one $w \in \mathcal{W}$. Therefore,

$$\begin{aligned} &\mathbb{P}(\text{AdaRank inverts } (j, k)) \\ &\leq \sum_{w \in \mathcal{W}} \mathbb{P}(\text{the fixed-}w \text{ score inverts } (j, k)), \end{aligned}$$

which gives the first bound by applying the fixed-weight result to each $w \in \mathcal{W}$. The second bound follows by upper bounding the sum by the largest term times $|\mathcal{W}|$. $\square$

D Experimental Details

This section records the construction details omitted from the main experiment setup.

Machine configuration. All experiments were run on a CPU workstation with an Intel Core Ultra 9 285HX CPU, 24 cores, and 128 GiB RAM. The implementation is in Python, and the real-data valuation utilities use scikit-learn [4]. No GPU acceleration is used. For runtime comparisons, the compared methods are executed under the same CPU and thread settings.

Dataset construction. In the main shared-ranking benchmark, SOUG, Breast Cancer, and Wine use all nine target semivalues. For SOUG, we use $n \in \{50, 100, 200, 500\}$, set $M = 5n$, draw each unanimity-set size uniformly from $\{2, 3, 4, 5, 6\}$, and draw nonnegative weights independently from $\text{Unif}(0, 1)$. We then normalize the weights so that $v(\mathcal{N}) = 100$. This bounded-size choice keeps nontrivial multi-player interactions while avoiding the near-zero WBZ signals that arise when low- q weighted Banzhaf values are applied to large unanimity sets. For Breast Cancer [5], we first hold out 99 examples as the test set, then sample a fixed player set of size $n \in \{50, 100, 200\}$ from the remaining training examples; the five sampling runs share this player set. Coalition utility is the test accuracy of a standardized SVM classifier [2, 4] trained on the coalition. For Wine [1], we first hold out 50 stratified examples as the validation set, then sample a fixed stratified player set of size $n \in \{50, 75, 128\}$ from the remaining examples. Coalition utility is the validation accuracy of the same standardized SVM pipeline.

Reference creation. SOUG reference rankings are computed exactly using the closed form in Section C.2. For Breast Cancer and Wine, exact semivalue computation is infeasible, so reference rankings are generated offline with high-budget WPERM sampling. For each setting, we increase the WPERM budget until every target semivalue passes the independent-replica stability criterion: minimum pairwise Spearman correlation at least 0.90 and Kendall correlation at least 0.75. The accepted references use three independent replicas for each setting, with larger budgets for harder instances. These references are used only as evaluation targets.

E Additional Experimental Results

This section reports additional results that support the main experimental evaluation. We first give experimental-design checks and additional ranking diagnostics, then report single-target comparisons, method-behavior analyses, runtime overhead, pairwise-inversion decay, and downstream robustness checks.

E.1 Sampling-Distribution Sensitivity

We compare three coalition-size sampling distributions in representative shared multi-semivalue settings: $q_s \propto 1/\sqrt{s(n-s)}$, OFA-S-Set, and $q_s \propto (s(n-s))^{0.5}$. OFA-S-Set is the set-averaged version of the single-target OFA-S variance-proxy rule [3]. Let $a_s^{(k)} = \binom{n-1}{s-1} p_s^{(k)}$ for target semivalue k . On the sampled interior sizes $s \in \mathcal{S}_{\text{rand}}$, we use

$$q_s^{\text{OFA-S-Set}} = \frac{\sqrt{\frac{1}{K} \sum_{k=1}^K \left[\frac{(a_s^{(k)})^2}{s} + \frac{(a_{s+1}^{(k)})^2}{n-s} \right]}}{\sum_{r \in \mathcal{S}_{\text{rand}}} \sqrt{\frac{1}{K} \sum_{k=1}^K \left[\frac{(a_r^{(k)})^2}{r} + \frac{(a_{r+1}^{(k)})^2}{n-r} \right]}}.$$

The comparison includes **IncRank**, **ExcRank**, **StdRank**, **AdaRank**, and **OFA**; the table reports **AdaRank** because it is the main proposed method. The table reports Spearman- ρ Log-AURC on SOUG and Breast Cancer with $n = 100$, and Wine with $n = 75$. The $q_s \propto 1/\sqrt{s(n-s)}$ distribution gives the best average performance in all three settings and the best worst-target performance on SOUG and Breast Cancer; OFA-S-Set is slightly better on Wine worst-target performance. The comparison also found $q_s \propto 1/\sqrt{s(n-s)}$ to be the strongest average-performing choice for the other included methods in these representative settings.

Table 2: q_s sensitivity for AdaRank. Entries are Spearman- ρ Log-AURC mean (std), with standard deviations over repeated sampling runs. Worst is the minimum over target semivalues for each setting and q_s . Higher is better.

Setting	q_s	Avg.	Worst
Breast Cancer	$1/\sqrt{s(n-s)}$	0.534 (0.006)	0.432 (0.017)
Breast Cancer	OFA-S-Set	0.509 (0.028)	0.407 (0.063)
Breast Cancer	$(s(n-s))^{0.5}$	0.490 (0.030)	0.342 (0.054)
SOUG	$1/\sqrt{s(n-s)}$	0.434 (0.028)	0.043 (0.065)
SOUG	OFA-S-Set	0.426 (0.027)	0.020 (0.071)
SOUG	$(s(n-s))^{0.5}$	0.310 (0.054)	-0.073 (0.073)
Wine	$1/\sqrt{s(n-s)}$	0.574 (0.027)	0.338 (0.127)
Wine	OFA-S-Set	0.512 (0.017)	0.348 (0.086)
Wine	$(s(n-s))^{0.5}$	0.554 (0.022)	0.294 (0.074)

E.2 Top-Selection Diagnostics

We report NDCG@20% as a supporting diagnostic for top-ranked data selection. For a setting with n players, this metric evaluates the top $\lceil 0.2n \rceil$ positions using the reference semivalue scores shifted to be nonnegative.

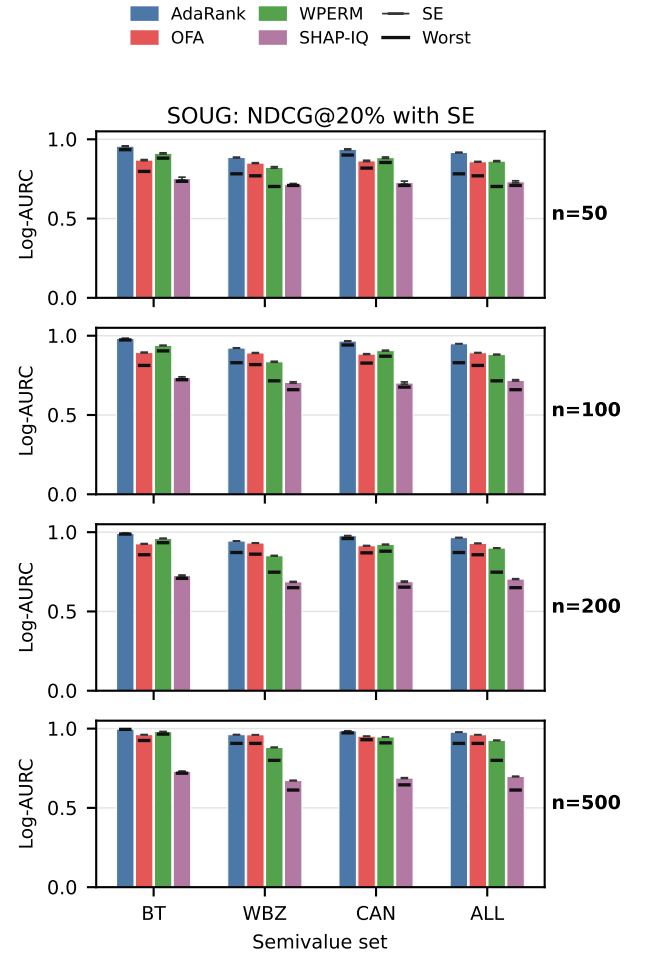


Figure 1: Top-ranked-data diagnostic on SOUG using NDCG@20% Log-AURC in the same shared multi-semivalue ranking setting as the SOUG ranking results in the main text. The plotting convention matches the main figures: bars show family averages with standard-error bars, and black ticks show the worst target in the family.

Figure 1 shows that **AdaRank** remains strongest on SOUG under the top-selection diagnostic by All-9 average point estimate for every tested n .

Figure 2a is also consistent with the main Breast Cancer result for $n = 100$ and $n = 200$. For $n = 50$, WPERM slightly exceeds **AdaRank** on this All-9 NDCG@20% diagnostic. Figure 2b gives the corresponding Wine result: **AdaRank** has the best All-9 average NDCG@20% Log-AURC for all three plotted n values.

E.3 Single-Target Ranking Results

We also evaluate a single-target setting, where each run is optimized for one specified semivalue. This setting allows comparison with single-target baselines such as GELS and WSL. It is useful as a

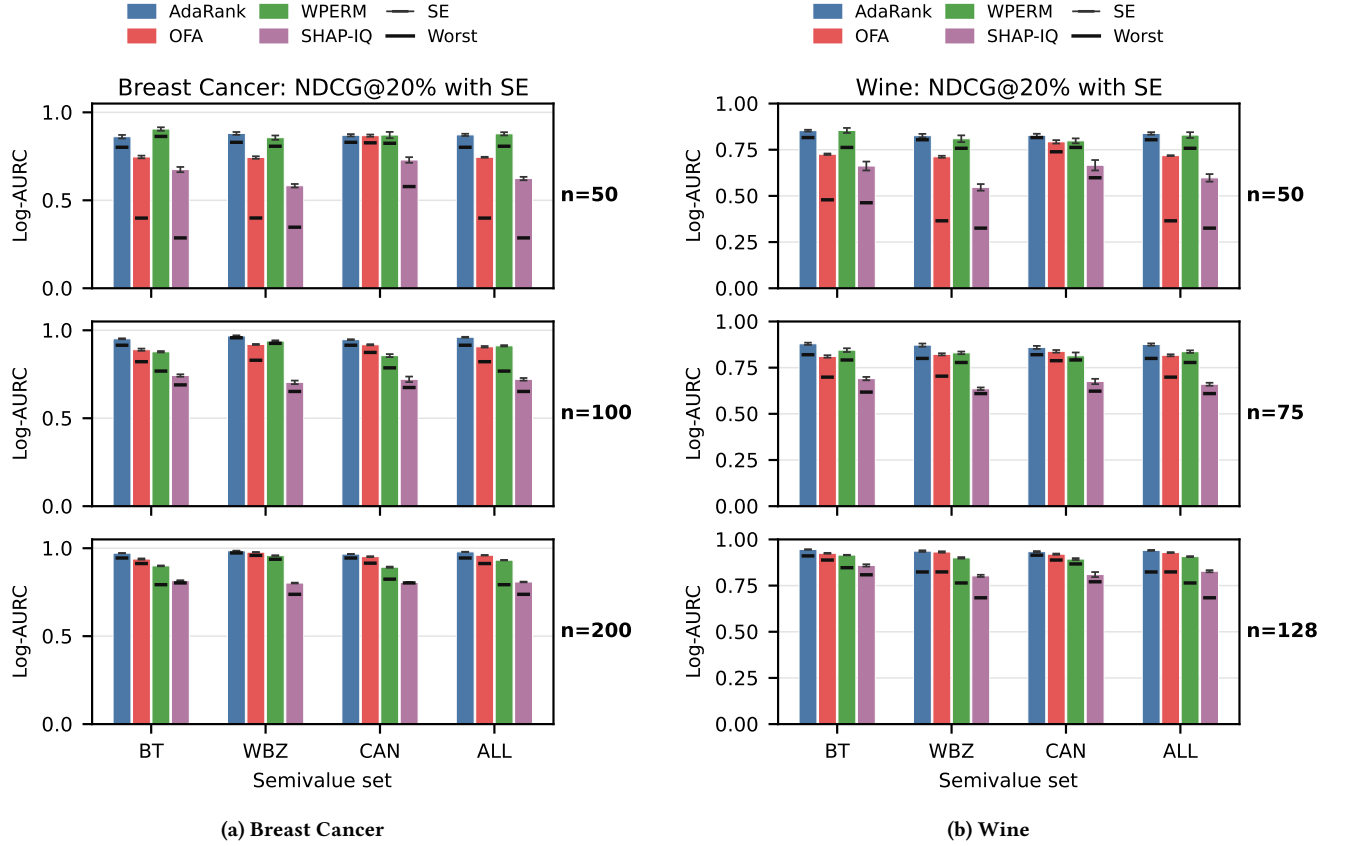


Figure 2: Top-ranked-data diagnostic on real-world data valuation benchmarks using NDCG@20% Log-AURC in the same primary shared multi-semivalue ranking setting as the Breast Cancer and Wine ranking results in the main text. The plotting convention matches Figure 1.

Table 3: Single target-semivalue ranking results. Entries are mean (std) Spearman- ρ Log-AURC over budgets $0.2n, \dots, 2n$, averaged over all nine target semivalues and tested sizes, and rounded to three decimals. SOUG and Breast Cancer use $n \in \{100, 200\}$, and Wine uses $n \in \{75, 128\}$. Higher is better. Bold marks the best method in each column and $\dagger$ marks the second-best method. Target-specific methods are diagnostic single-target references.

Method	SOUG	Breast Cancer	Wine
AdaRank	0.942 (0.013)	0.592 (0.020) $\dagger$	0.641 (0.023) $\dagger$
WSL	0.896 (0.016)	0.502 (0.055)	0.618 (0.030)
OFA-S	0.910 (0.019) $\dagger$	0.553 (0.030)	0.602 (0.021)
WPERM	0.770 (0.027)	0.489 (0.020)	0.496 (0.031)
GELS	0.640 (0.020)	0.672 (0.015)	0.693 (0.026)
SHAP-IQ	0.263 (0.026)	0.056 (0.018)	0.056 (0.028)

separate performance evaluation, but it is not the shared multi-semivalue ranking task because those baselines may use separate sampling distributions for different semivalues.

Table 3 shows that **AdaRank** remains competitive even in this single-target setting. On SOUG, **AdaRank** gives the best average

Log-AURC, while OFA-S is strongest among the external single-target baselines. On Breast Cancer and Wine, GELS achieves the best average performance, reflecting the advantage of single-target sampling. We therefore use OFA-S for SOUG and GELS for Breast Cancer and Wine as the split-budget baselines in the main-text comparison, giving the split-budget strategy a favorable single-target configuration. **AdaRank** is nevertheless second-best on average for both real-data benchmarks and outperforms OFA-S, WPERM, and SHAP-IQ on the aggregate average. Thus, this single-target evaluation supports the robustness of **AdaRank**.

E.4 Adaptive Weight Selection

We next inspect the weights selected by **AdaRank**. In **AdaRank**, $w = 1$ means the method uses the standardized **IncRank** score, $w = 0$ means it uses the standardized **ExcRank** score, and intermediate values interpolate between the two. A boundary value such as $w = 1$ is interpreted as an internal stability decision: among the candidate mixtures, the include-side ranking was the most reproducible across the two folds of the same sampled coalitions.

Figure 3a shows that the selected weights are highly semivalue dependent on SOUG. BT-1-1 and BT-1-4 concentrate near $w = 1$, so

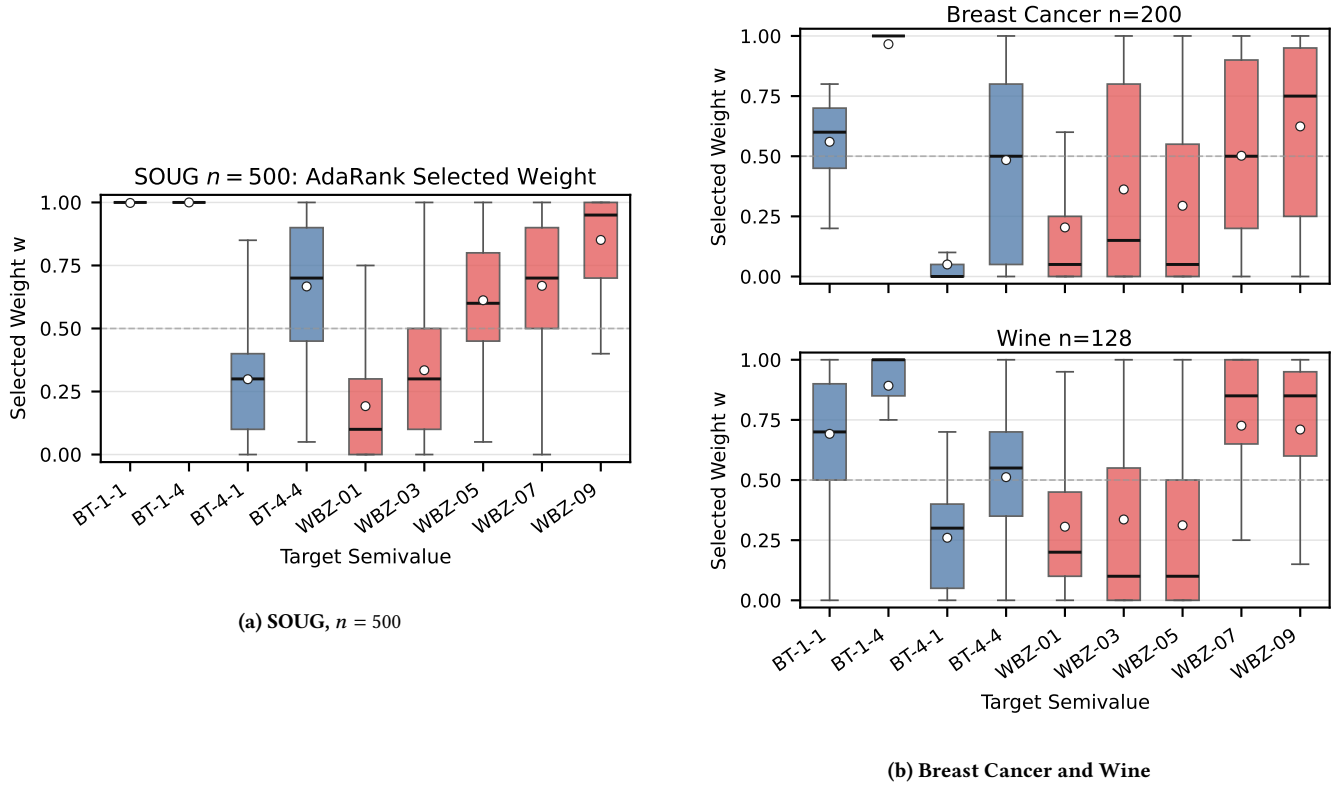


Figure 3: Selected AdaRank weights on SOUG and real-world data valuation benchmarks. A value $w = 1$ corresponds to selecting the standardized IncRank signal within AdaRank’s same-sample split-fold criterion.

AdaRank effectively uses **IncRank** for these targets. In contrast, BT-4-1 and WBZ-01 place much smaller weights on **IncRank**. This is consistent with the coverage difference between the two proxies: a size- s sample updates **IncRank** for s included players but **ExcRank** for $n - s$ excluded players. Targets emphasizing smaller coalitions therefore give **ExcRank** more stable per-player evidence, while larger-coalition targets tend to favor **IncRank**.

Figure 3b gives the corresponding view for the two real-world data valuation tasks. The weights are more dispersed than on SOUG, but the same pattern remains: BT-1-4 favors large w , while BT-4-1 and low weighted-Banzhaf targets favor smaller w . As a post-hoc check, we compare the direction of the selected weight with the better component in hindsight. Across the six real-data dataset-size settings and nine semivalues, $\bar{w} > 0.5$ agrees with $\rho(\text{IncRank}) > \rho(\text{ExcRank})$ in 49 of 54 cases. The five disagreements are negligible: all have absolute Spearman- ρ gaps below 0.006. When the selected weight is decisive, meaning $\bar{w} \geq 0.7$ or $\bar{w} \leq 0.3$, this agreement holds in all 31 cases.

E.5 Runtime Overhead

AdaRank uses the same sampled coalition utilities as **IncRank** and **ExcRank**, so it does not increase the utility-evaluation budget. Its additional wall-clock cost comes from maintaining split-fold statistics and selecting the mixture weight over a finite grid. We

Table 4: Wall-clock runtime overhead of AdaRank relative to IncRank after discarding one warm-up timing task per setting. Entries report mean (std) of run-level runtime ratios, AdaRank/IncRank, at each budget factor. Lower is better; 1.0 means equal runtime.

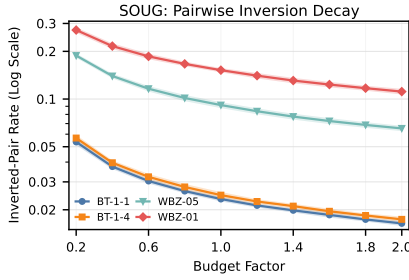
Setting	n	$0.2n$	$0.4n$	$0.6n$	$1.0n$
SOUG	50	5.62 (0.071)	5.49 (0.062)	5.44 (0.077)	4.83 (0.057)
SOUG	500	1.41 (0.004)	1.41 (0.003)	1.41 (0.003)	1.40 (0.003)
Breast Cancer (SVM)	50	1.09 (0.001)	1.09 (0.000)	1.09 (0.000)	1.07 (0.001)
Breast Cancer (SVM)	200	1.02 (0.002)	1.02 (0.002)	1.02 (0.003)	1.02 (0.005)

therefore measure the runtime ratio **AdaRank**/IncRank under the same sampling budget and fixed CPU/thread settings.

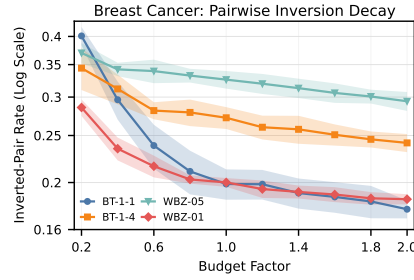
Table 4 shows that this overhead depends strongly on the cost of the underlying utility evaluations. On SOUG with $n = 50$, each utility evaluation is extremely cheap, so this row is best interpreted as a microbenchmark of **AdaRank**’s fixed post-processing cost; the ratio is around 5, but the absolute runtimes are only milliseconds. At $n = 500$, coalition scoring is more expensive and the fixed split-fold/grid-selection cost is amortized, reducing the ratio to about 1.4. The effect is clearer on Breast Cancer, where each utility evaluation fits an SVM: the overhead is close to one for $n = 50$, about 1.07–1.09, and about 1.02 for $n = 200$. Thus **AdaRank**’s adaptive weighting adds little wall-clock overhead when utility evaluation dominates.

Table 5: CIFAR-10N audit-size sensitivity for AdaRank at per-player budget $b = 100$. Best and worst single-semivalue rows are selected by model delta separately for each audit size, and Mean Single averages over the nine individual target choices. Entries are mean (std) over three runs; higher is better except for the number of flagged examples.

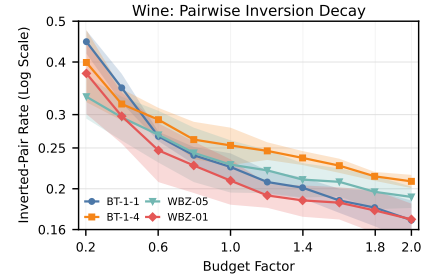
Audit size	Policy	# flagged	Precision	Recall	Lift	Model delta
$k = 0.05n$	Best single (BT-1-4)	100.0 (0.0)	0.200 (0.010)	0.112 (0.006)	2.2 (0.1)	+0.073 (0.005)
$k = 0.05n$	Mean single	100.0 (0.0)	0.209 (0.012)	0.117 (0.007)	2.3 (0.1)	+0.054 (0.002)
$k = 0.05n$	Worst single (WBZ-01)	100.0 (0.0)	0.237 (0.031)	0.133 (0.017)	2.7 (0.3)	+0.032 (0.004)
$k = 0.05n$	All-9 intersection	5.0 (1.0)	0.467 (0.058)	0.013 (0.003)	5.2 (0.6)	+0.005 (0.000)
$k = 0.05n$	All-9 union	366.7 (3.8)	0.163 (0.008)	0.335 (0.018)	1.8 (0.1)	+0.117 (0.010)
$k = 0.15n$	Best single (BT-4-4)	300.0 (0.0)	0.188 (0.016)	0.316 (0.027)	2.1 (0.2)	+0.126 (0.004)
$k = 0.15n$	Mean single	300.0 (0.0)	0.169 (0.003)	0.284 (0.006)	1.9 (0.0)	+0.097 (0.004)
$k = 0.15n$	Worst single (WBZ-01)	300.0 (0.0)	0.178 (0.011)	0.300 (0.018)	2.0 (0.1)	+0.061 (0.004)
$k = 0.15n$	All-9 intersection	22.3 (1.2)	0.343 (0.087)	0.043 (0.012)	3.9 (1.0)	+0.021 (0.004)
$k = 0.15n$	All-9 union	895.0 (5.3)	0.124 (0.005)	0.625 (0.029)	1.4 (0.1)	+0.136 (0.008)



(a) SOUG, $n = 500$



(b) Breast Cancer, $n = 200$



(c) Wine, $n = 128$

Figure 4: Empirical pairwise-inversion decay for AdaRank. Curves show mean inverted-pair rates for representative semivalue targets over per-player budgets from $0.2n$ through $2n$; bands show one standard deviation and the y-axis is logarithmic.

E.6 CIFAR-10N Audit-Size Sensitivity

Table 5 reports the same CIFAR-10N workflow diagnostic as the main paper for two additional audit sizes. The same removal action is used for every policy, and model delta is always the change in clean validation accuracy after removing the selected examples and retraining. The All-9 intersection consistently gives the highest precision/lift but selects very few examples, while the All-9 union gives the largest model improvement at both audit sizes.

E.7 Pairwise Inversion Decay

We measure the inverted-pair rate: the fraction of unordered player pairs whose order under **AdaRank** disagrees with the reference ranking. Figure 4 reports SOUG with $n = 500$, Breast Cancer with $n = 200$, and Wine with $n = 128$, for four representative semivalues over budgets from $0.2n$ through $2n$. The curves generally decline; their near-linear shape on the logarithmic y-axis, especially on

SOUG, suggests approximately exponential empirical decay. The real-data curves are noisier because their references are approximate and each utility evaluation retrains a model.

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